A Project Report

On

GAIN ENHANCEMENT TECHNIQUES FOR PRINTED EMBEDDED GNSS ANTENNA

BY

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Thank you.



Birla Institute of Technology and Science-Pilani,

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Certificate

This is to certify that the project report entitled "**GAIN ENHANCEMENT TECHNIQUES FOR PRINTED EMBEDDED GNSS ANTENNA**" submitted by Mr. Venkat Aadith Reddy Madderla (ID No.2023A3PS0302H) in complete fulfillment of the requirements of the course EEE F377, Design Project Course, embodies the work done by him under my supervision and guidance.

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ABSTRACT

As wireless communication systems rapidly evolve toward miniaturization, integrating reliable positioning capabilities within constrained footprints remains a critical RF design challenge. Small-form-factor Internet of Things (IoT) devices increasingly rely on Global Navigation Satellite Systems (GNSS), yet compact printed antennas inherently suffer from degraded gain and radiation efficiency. This project focuses on the design, simulation, and analysis of a compact printed microstrip patch antenna explicitly targeted for the GNSS L1 frequency band (1.575 GHz). Utilizing CST Studio Suite, the antenna was modeled on a cost-effective FR-4 substrate.

To overcome the inherent gain limitations typical of miniaturized patch antennas, this design leverages a metamaterial-inspired structure, initially validated using a single Split Ring Resonator (SRR) configured as a superstrate. By suspending this SRR layer at an optimized height directly above the radiating patch, it functions as an electromagnetic lens. This parasitic interaction modifies the near-field phase distribution, focusing the radiated energy to significantly improve broadside directivity without expanding the antenna's primary physical footprint.

Following the successful proof of concept, the design was scaled into periodic 2x2 and 4x4 SRR unit cell arrays to act as a distributed metamaterial lens. This transition allows for precise control over the phase and amplitude of near-field emissions, transforming divergent spherical wave fronts into uniform planar waves. By artificially enlarging the antenna's physical aperture, the periodic pattern captures peripheral near-field energy that would typically be lost to side lobes or surface waves, redirecting it forward to maximize radiation efficiency.

Simulation results validated the effectiveness of this hierarchical approach. The baseline patch antenna demonstrated a peak gain of 1.93 dB with a return loss of -22.41 dB. The integration of a single SRR elevated the gain to 2.26 dB. Through the implementation of the periodic metasurface, performance was further optimized, with the 2x2 array achieving 2.45 dB and the 4x4 array reaching a final peak gain of 2.65 dB. Throughout these enhancements, the antenna maintained excellent impedance matching, with the 4x4 configuration exhibiting a return loss of -20.05 dB. This work confirms that metamaterial-inspired periodic superstrates are a highly viable strategy for developing high-gain, compact antennas on standard, low-cost substrates.

CONTENTS

Title page……..…...1

Acknowledgements……...2

Certificate…….....3

Abstract…….....4

1. Introduction …….…….....6
    1.1 Overview of GNSS Technology…….6
    1.2 Challenges of Embedded GNSS Antenna……...6
    1.3 Project Scope and Objective……7
    1.4 Different bands of GNSS Antenna…….....7

2. Theoretical Background……8
    2.1 Microstrip Patch Antenna……8
    2.2 Key Design Parameters……8

3. Methodology……10
    3.1 Simulation Software……10
    3.2 Design Approach……10
    3.3 Rationale for Performance Enhancement Techniques…….....11

4. Implementation……..13
    4.1 Design Parameters and Specifications……13

5. Results and Analysis……..15
    5.1 Comparision of the Basic Antenna with the SRR Gain Enhanced Antenna……15
    5.2 Results of Antenna with SRR configured as Superstrate……16
    5.3 Comparison of the results of the designs……..17
    5.4 Comparative Analysis between the Antenna Designs……..17

POST-MIDSEM……18

6. Execution of Phase-2…….....19
    6.1 Scaling to Metasurface Array……..19
    6.2 Array Optimization and Tuning……...19

7. Results and Analysis……..20
    7.1 2X2 SRR Unit Cell Array……20
    7.2 4X4 SRR Unit Cell Array……21
    7.3 Comparision of Results……23
    7.4 Electromagnetic Mechanisms of 4X4 Array……23

Conclusion……..24

References……...25

INTRODUCTION

An Embedded Global Navigation Satellite System (GNSS) antenna is a compact, miniaturized antenna designed to be integrated directly within an electronic device. Unlike their external counterparts, which are mounted on the outside of a device, embedded antennas are concealed within the product's housing, offering a seamless and aesthetically pleasing design. This integration is crucial for the functionality of a vast array of modern portable electronics.

Overview of GNSS Technology

A Global Navigation Satellite System (GNSS) antenna is a specialized transducer designed to receive radio frequency signals transmitted by various satellite constellations. It serves as the primary interface between the satellite signals in space and the processing hardware of a GNSS receiver. The functionality of a modern GNSS antenna extends beyond a single system, enabling it to capture signals from multiple constellations such as the United States' Global Positioning System (GPS), Russia's GLONASS, the European Union's Galileo, and China's BeiDou. This multi-constellation capability significantly enhances the availability, accuracy, and integrity of positioning, navigation, and timing (PNT) solutions, particularly in challenging signal environments like urban canyons or areas with dense foliage.

The core function of a GNSS antenna is the conversion of electromagnetic waves from satellites into electrical signals that can be interpreted by a receiver. The faint signals, having travelled thousands of kilometres from space, are captured by the antenna's radiating element. This interaction induces a small voltage, creating an electrical current that mirrors the properties of the incoming satellite signal.

Challenges of Embedded GNSS Antenna

The integration of Global Navigation Satellite Systems (GNSS) into severely space-constrained devices presents a fundamental electromagnetic challenge. As the physical dimensions of a printed antenna are scaled down to meet the strict form-factor requirements of embedded systems, its radiation characteristics inherently degrade. The primary bottleneck in this miniaturization process is the severe penalty inflicted on antenna gain. In electrically small antennas, the reduction in effective radiating area directly diminishes both directivity and radiation efficiency. This degradation is governed by the fundamental limits of radiation; as the physical volume decreases, the antenna's quality factor inherently increases, complicating impedance matching and trapping more energy in the near-field. Furthermore, the use of cost-effective commercial dielectric substrates often exacerbates these issues. The higher loss tangents of such materials dissipate valuable RF energy as heat, while the compact geometry tends to excite unwanted surface waves that remain trapped within the substrate rather than radiating into free space. Cumulatively, these factors translate to an unacceptably low overall broadside gain, which critically compromises the link margin required for reliable and continuous satellite signal acquisition. Consequently, overcoming these physical limitations to design high-gain, compact embedded antennas without expanding their physical footprint remains a pressing engineering imperative for robust localization.

Project Scope and Objective

This project directly addresses these fundamental challenges by designing a compact, embedded microstrip patch antenna optimized for superior radiation performance. The initial phase of this research focuses on developing and simulating a base antenna on a cost-effective FR-4 substrate, specifically targeted for the GNSS L1 frequency band (1.575 GHz) using CST Studio Suite. To explicitly overcome the low-gain bottleneck typical of miniaturized designs and enhance overall performance, a metamaterial-inspired structure is being introduced. Specifically, a single Split Ring Resonator (SRR) has been designed to act as a superstrate placed directly above the radiating patch. By functioning as a parasitic director, this superstrate configuration aims to improve directivity and radiation efficiency without increasing the primary footprint of the antenna. This report details the design of the base antenna parameters and introduces the preliminary geometric modeling of the SRR superstrate. Finally, it outlines the future work required to fully simulate, integrate, and optimize the superstrate configuration to validate the expected gain enhancements prior to physical fabrication.

Different Bands of the GNSS Antenna

System	**Band**	**Center Frequency (MHz)**	Applications
GPS	L1	1575.42	Civilian band (C/A code)
GPS	L2	1227.60	Military/dual-frequency
GPS	L5	1176.45	Safety-of-life, aviation
GLONASS	L1	~1602	Similar to GPS L1
Galileo	E1	1575.42	Compatible with GPS L1
Galileo	E5	1191.795	Similar to GPS L5
BeiDou	B1	1561.098	Regional & global signals

THEORETICAL BACKGROUND

Microstrip Patch Antenna

A microstrip patch antenna is a low-profile antenna widely used in wireless and satellite communication systems due to its compact size, ease of fabrication, and compatibility with printed circuit technology. It consists of a thin metallic patch printed on a dielectric substrate with a continuous ground plane on the opposite side. The patch can take different shapes such as rectangular, circular, or triangular, though the rectangular form is most common because of its simple analysis and fabrication. The antenna operates primarily by fringing fields at the edges of the patch, radiating electromagnetic waves when excited at a resonant frequency. The antenna's resonant frequency and other performance characteristics are highly dependent on the physical dimensions of the patch, particularly its length and width, as well as the properties of the substrate material, specifically its thickness (h) and dielectric constant (ε_r). By carefully selecting these parameters, the antenna can be designed to operate efficiently at a specific frequency.

Key Design Parameters

The design of a rectangular microstrip patch antenna begins with determining its physical dimensions to achieve resonance at a desired operating frequency (f_r). The width (W) of the patch is a critical parameter that influences the antenna's impedance and radiation pattern. It can be calculated using the following formula, where c is the speed of light:

$$W = \frac{c}{2f_r}\sqrt{\frac{2}{\varepsilon_r + 1}}$$

After calculating the patch width, you must determine its resonant length. This is complicated by the **fringing effect**, where electromagnetic fields at the patch's edges bulge into the air, making the antenna behave as if it's **electrically longer** than its physical dimensions. To compensate for this, you first need to find the **effective dielectric constant (ε_{eff})**. This value is essentially a weighted average that accounts for the fields travelling through both the substrate and the air. Because air has a much lower dielectric constant than the substrate, the resulting effective value (ε_{eff}) will always be less than the substrate's actual dielectric constant (ε_r).

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{W}\right)^{-\frac{1}{2}}$$

The fringing effect also causes an extension in the patch's length (ΔL). This length extension can be calculated with the formula:

$$\frac{\Delta L}{h} = 0.412\frac{\left(\varepsilon_{eff} + 0.3\right)\left(\frac{W}{h} + 0.264\right)}{\left(\varepsilon_{eff} - 0.258\right)\left(\frac{W}{h} + 0.8\right)}$$

The actual length of the patch is then determined by subtracting twice this extension from the half-wavelength resonant length, calculated using the effective dielectric constant, where L is the actual length of the patch:

$$L_{eff} = \frac{c}{2f_r\sqrt{\varepsilon_{eff}}}$$

$$L = L_{eff} - 2\Delta L$$

To improve performance, techniques such as using thicker substrates, adding parasitic patches, or employing array configurations are often used. Despite their limitations, microstrip patch antennas remain one of the most versatile and widely used types in modern communication and GNSS applications.

METHODOLOGY

This section outlines the systematic approach taken for the design, simulation, and analysis of the embedded GNSS antenna. The methodology is structured to first validate our simulation environment against an established design before proceeding to the development of our proposed antenna.

Simulation Software: CST Studio Suite

The entire design and simulation process for this project was conducted using CST Studio Suite. This software was selected as it is an industry-standard tool renowned for its high accuracy and efficiency in 3D electromagnetic (EM) field simulation. Unlike standard FEM-only tools, CST operates primarily on the Finite Integration Technique (FIT) within its Time Domain Solver, alongside the Finite Element Method (FEM) in its Frequency Domain Solver, to rigorously solve Maxwell's equations and provide a precise analysis of high-frequency components like antennas.

Our methodology within the CST environment involved several key steps. First, a detailed 3D model of the microstrip patch antenna was constructed, specifying the exact geometry of the patch, substrate, and ground plane. Following this, material properties were assigned, specifically defining the dielectric constant (ε_r) and thickness of the chosen FR-4 substrate.

Through the transient simulation, we were able to extract critical performance metrics for the antenna. These included the S-parameters (specifically S_{11} or return loss), Voltage Standing Wave Ratio (VSWR), input impedance, the 3D far-field radiation pattern, and antenna gain. This simulation-driven approach enabled a process of virtual prototyping, allowing us to iteratively adjust the antenna's dimensions using the built-in parameter sweep functions to optimize its performance for the GNSS L1 frequency band without the need for costly and time-consuming physical fabrication.

Design Approach

Our design process was divided into two primary stages:

Stage 1: Theoretical Design and Parametric Tuning of the Baseline Antenna: The project commenced with the analytical design of an "ideal" rectangular microstrip patch antenna. The initial physical dimensions, primarily the patch width and length, were calculated using standard theoretical formulas for microstrip antennas. These baseline calculations were derived using the dielectric properties of the selected FR-4 substrate and the target resonant frequency of the GNSS L1 band (1.575 GHz). This mathematically calculated geometry was then modeled in CST Studio Suite to establish a starting point. However, because standard theoretical formulas provide approximations that do not fully account for complex fringing fields and precise substrate interactions, parametric tuning was essential. The dimensions of the patch and the feed structure were iteratively adjusted and optimized within the simulation environment. This fine-tuning process allowed us to precisely lock the resonant frequency exactly at 1.575 GHz and achieve an optimal return loss.

Stage 2: Development of the Metamaterial Superstrate for Gain Enhancement: Building upon the fully tuned baseline design from Stage 1, the second stage focuses on overcoming the fundamental low-gain limitations inherent to the compact FR-4 patch. The objective is to explicitly enhance the radiation performance for the L1 band without expanding the primary footprint of the antenna. Drawing inspiration from the gain enhancement techniques demonstrated by Choudhary et al. [1], a metamaterial-inspired approach is being developed. This involves the design of a single Split Ring Resonator (SRR) configured to function as a superstrate layer, placed strategically above the radiating patch. As validated in [1], utilizing a metamaterial cell-based superstrate allows the structure to act as a parasitic director. This configuration is engineered to focus the radiated electromagnetic energy, thereby significantly enhancing the overall directivity and peak gain of the antenna system.

Rationale for Performance Enhancement Techniques

To overcome the inherent low-gain limitations of compact printed antennas on standard FR-4 substrates, the proposed custom design incorporates a metamaterial-inspired Split Ring Resonator (SRR) configured as a superstrate. In conventional miniaturized patch antennas, the radiated energy typically suffers from a broad beamwidth and poor directivity, resulting in low overall gain. By suspending a specifically tuned SRR structure at an optimized height directly above the radiating patch, the SRR functions as a parasitic director or an electromagnetic lens. Electromagnetic coupling between the primary radiating patch and the superstrate modifies the phase distribution of the near-field emissions. This interaction focuses the radiated energy into a narrower, more directed beam, significantly enhancing the broadside directivity and overall peak gain without expanding the physical footprint of the primary antenna base.

The implementation involved designing a single metallic split ring on a secondary dielectric layer. To maximize the gain enhancement explicitly at the target GNSS L1 band (1.575 GHz) without detuning the base antenna's fundamental resonance, a rigorous parametric sweep was conducted using CST Studio Suite. The optimization process focused on fine-tuning critical geometric variables, including the SRR's radius, trace width, and split gap dimensions, as well as the crucial air gap spacing between the primary patch and the superstrate. This tuning ensured optimal electromagnetic coupling and maximum broadside radiation while maintaining excellent impedance matching at the feed point.

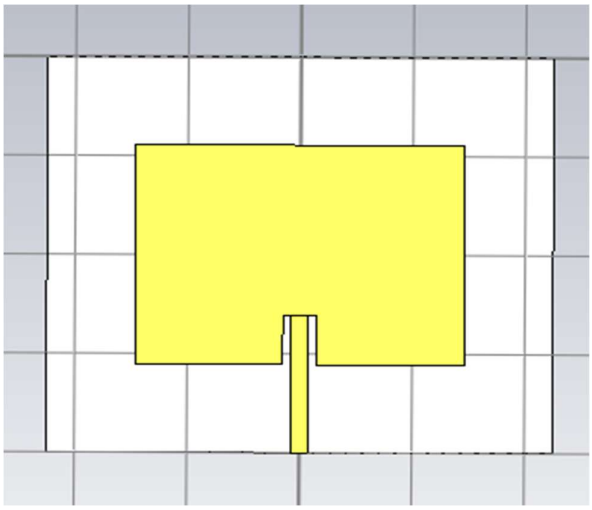


Fig.3.1: Initial Microstrip Patch Antenna Design

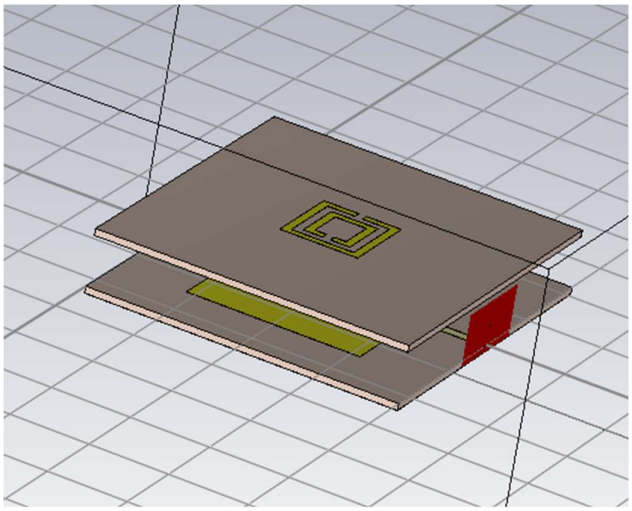


Fig.3.2: Proposed Antenna Design

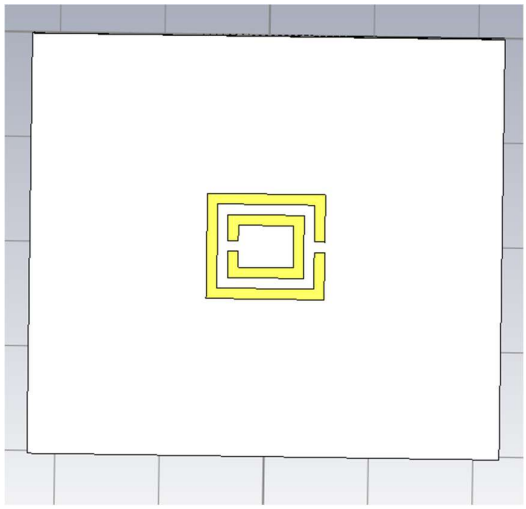


Fig.3.3: Top Superstrate Layer with SRR

IMPLEMENTATION

The implementation phase of this project began by translating the theoretical design into a virtual model. The initial physical dimensions for the microstrip patch antenna were determined using the standard transmission line model formulas, which provided a foundational baseline for the design. Following these calculations, the antenna was constructed and simulated using CST Studio Suite. This powerful software allowed for the detailed analysis and iterative optimization of the initial design to meet the required performance specifications.

Design Parameters and Specifications

The initial phase of our implementation involved building an Microstrip Patch Antenna with inset feed using the standard formulae. By adhering precisely to the dimensions and specifications obtained from the formulae, we created a foundational model. This process served to validate our simulation setup and establish a crucial reference point for the development and evaluation of our own design. The specifications are detailed below:

Substrate	FR-4
Dielectric Constant (ε_r)	4.4
Patch Length (L)	44.4mm
Patch Width (W)	58.5mm
Feed Length	17.8mm
Feed Width	2.987mm
Thickness of Substrate (h)	1.6mm
Target Impedance	50Ω
Ground Length	80mm
Ground Width	90mm
Target Frequency	1.575GHz

Following the baseline analysis, the project's second phase involved the implementation of a custom-designed antenna. This design is based on a rectangular microstrip patch architecture with a Split Ring Resonator (SRR) configured as a superstrate. The specific parameters governing this custom design are as follows:

Substrate & Superstrate	FR-4
Dielectric Constant (ε_r)	4.4
Patch Length (L)	44.5mm
Patch Width (W)	58.5mm
Feed Length	17.75mm
Feed Width	2.987mm
Thickness of Substrate (h)	1.6mm
Target Impedance	50Ω
Ground Length	80mm
Ground Width	90mm
Target Frequency	1.575GHz
Height between the layers	12mm
Thickness of the Superstrate	1.6mm
Width of the SRR	2mm

The specifications for the Frequency Sweep are as follows:

Start Frequency	1.0 GHz
Stop Frequency	3.0 GHz

RESULTS AND ANALYSIS

Comparative Analysis of the Basic Antenna with the SRR Gain enhanced Antenna

Results of the Baseline Antenna

This section presents the simulated performance of the antenna designs discussed in the previous chapter. The results are presented in a comparative manner, beginning with the performance of the replicated reference antenna. This is followed by an analysis of our initial design and, finally, the optimized results achieved after incorporating the Split Ring Resonator (SRR) configured as a superstrate.

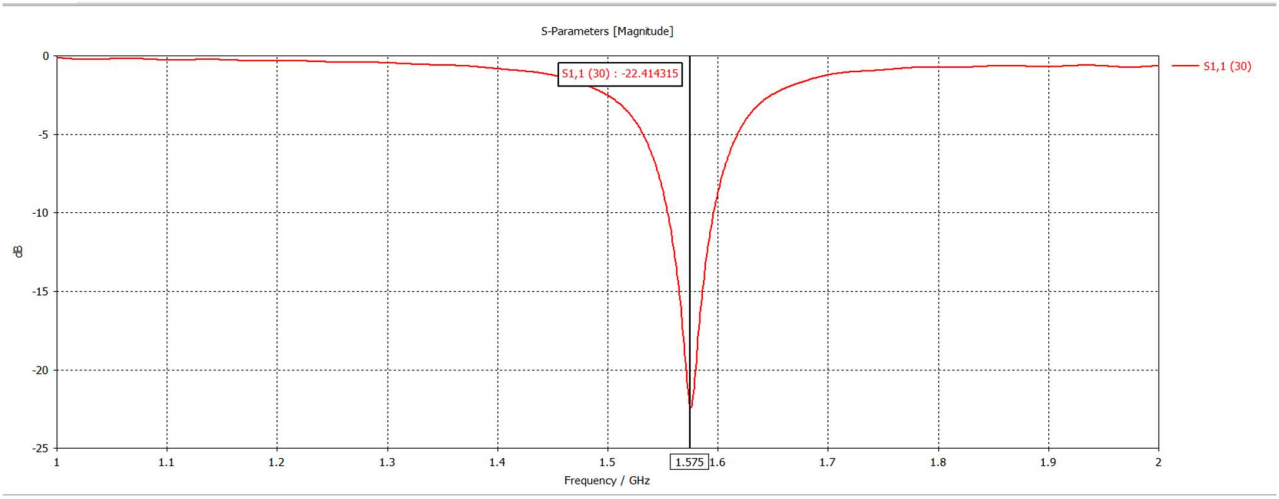


Fig.5.1 S_{11} Parameter for the Initial Basic Design. It was simulated between 1GHz to 2GHz. (S_{11} Parameter at 1.575GHz :- -22.414dB)

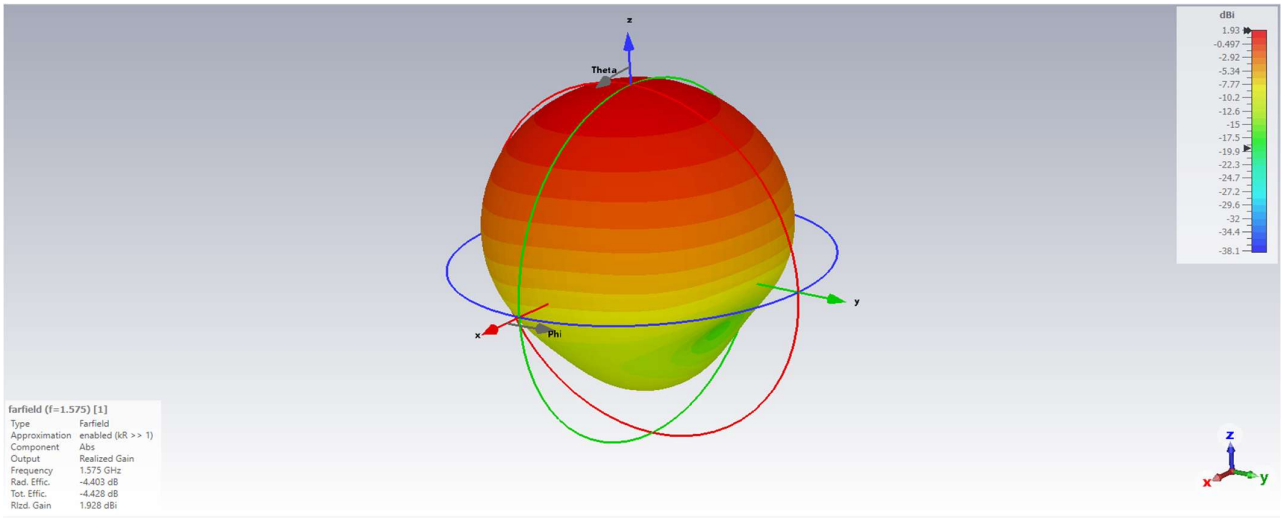


Fig.5.2 Gain for the initial Antenna

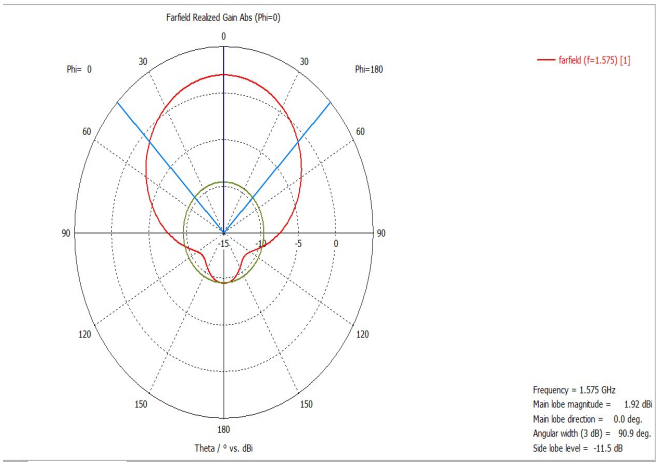


Fig.5.3: H-Plane

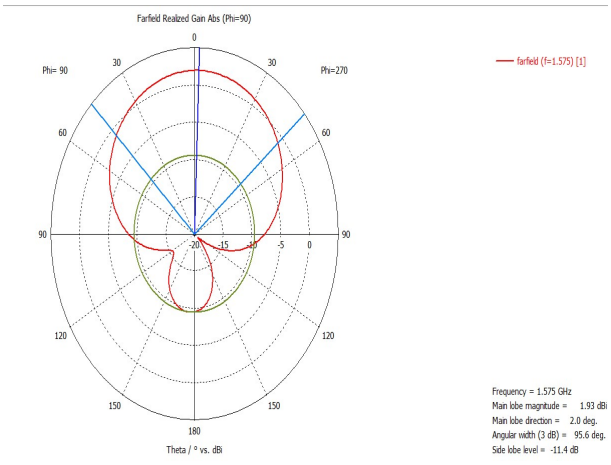


Fig.5.4: E-Plane

Results of Antenna with Split Ring Resonator (SRR) configured as a superstrate

The below graphs are the results for our design. We have compared the results of the design with and without the Split Ring Resonator (SRR) configured as a superstrate.

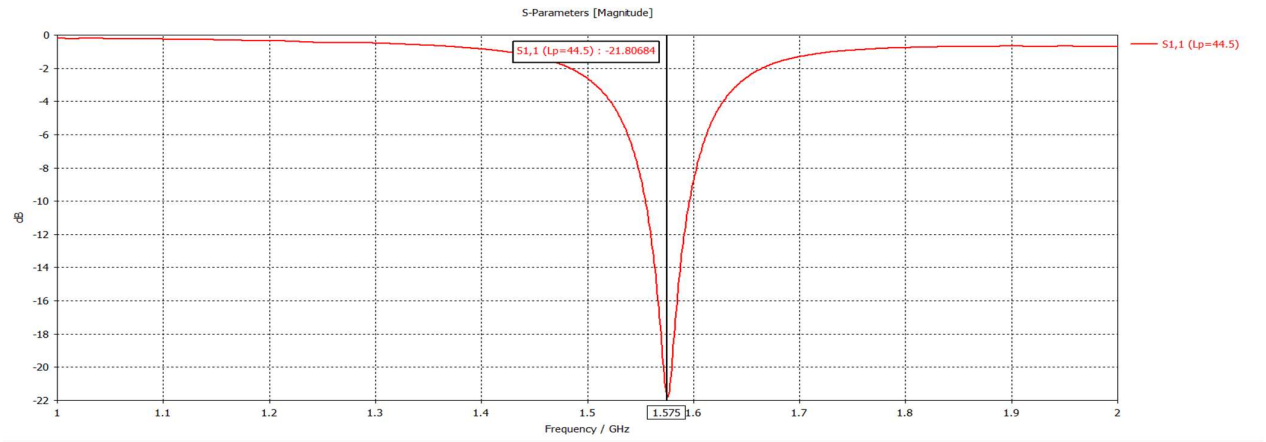


Fig.5.5 S_{11} Parameter for antenna with Split Ring Resonator (SRR) configured as a superstrate

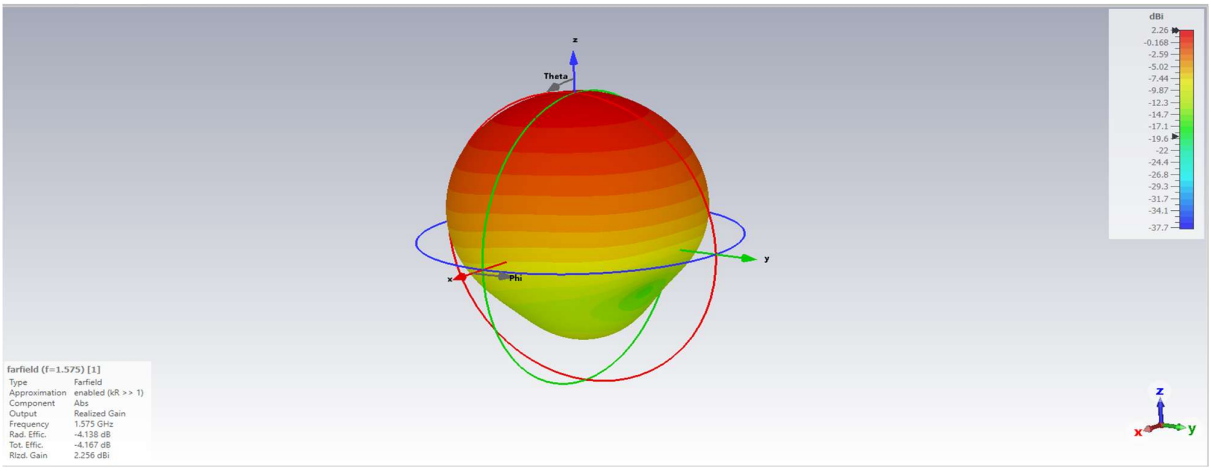


Fig.5.6 Gain for antenna with Split Ring Resonator (SRR) configured as a superstrate

Comparision of the results of the designs

	Basic Microstrip Patch Antenna	Microstrip Patch Antenna with a single Split Ring Resonator (SRR) configured as a superstrate
S_{11} Parameter	-22.414dB	-21.8dB
Realized Gain	1.93dB	2.26dB

Comparative Analysis between the Antenna Designs

This section presents a comparative analysis between the two simulated antenna configurations: the baseline basic microstrip patch antenna and the proposed design featuring a single Split Ring Resonator (SRR) configured as a superstrate. The primary metrics evaluated are the S_{11} Parameter and maximum Realized Gain, which collectively determine the antenna's operational effectiveness for GNSS L1 applications. This analysis serves to quantify the specific performance enhancements achieved through the implementation of the metamaterial-inspired superstrate.

An analysis of the S_{11} parameter, which indicates the quality of impedance matching, demonstrates that both designs maintain highly efficient power transfer at the target 1.575 GHz frequency. The basic microstrip patch antenna established a strong baseline with an excellent return loss of -22.414 dB. Upon introducing the SRR superstrate, the return loss shifted slightly to -21.8 dB. While this represents a marginal change in the absolute value, both figures remain well below the standard operational threshold of -10 dB. This indicates that the addition of the parasitic superstrate successfully preserved the antenna's fundamental resonance and excellent impedance matching without causing detrimental detuning.

An evaluation of the antenna gain, a measure of directivity and radiation efficiency, directly highlights the efficacy of the proposed design modifications. The basic microstrip patch antenna achieved a baseline peak gain of 1.93 dB, which is typical for a compact FR-4 design but can limit the link margin for satellite signal acquisition. However, the implementation of the SRR superstrate acting as an electromagnetic lens successfully focused the radiated energy, elevating the peak gain to 2.26 dB. This represents a clear gain enhancement of 0.33 dB over the baseline model. This tangible improvement confirms that the superstrate configuration effectively increases radiation directivity without expanding the antenna's primary footprint, successfully validating the core objective of the project.

POST-MIDSEM

Execution of Phase 2

Scaling to a Metasurface Array

Following the successful validation of the single-element superstrate, the project transitioned from a single parasitic director to a periodic metasurface array to maximize the effective aperture and radiation efficiency. This phase was implemented through the following steps:

- **Design Realization:** The theoretical N×N array was developed as a distributed metamaterial lens, iterating specifically through 2x2 and 4x4 SRR unit cell arrays.
- **The 2x2 Intermediate Proof:** Before moving to the full 4x4 grid, the 2x2 array served as a critical intermediate proof-of-concept. It was used to validate the project's core theory regarding the physics of scaling demonstrating that increasing the number of resonators provides more precise control over the phase and amplitude of near-field emissions.
- **Performance Bridge:** The 2x2 array successfully bridged the gap between the single superstrate and the final design, elevating the peak gain to 2.45 dB and maintaining an $S_{11\ of}$ -19.96 dB.
- **Aperture Expansion:** Scaling to these periodic arrays artificially enlarged the antenna's physical aperture, allowing it to capture peripheral near-field energy that would typically be lost to side lobes or surface waves.

Array Optimization and Tuning

To ensure the metasurface functioned efficiently without detuning the baseline antenna, several critical parameters were optimized post-midsem:

- **Periodicity Tuning:** Inter-element spacing between the split rings was adjusted to suppress unwanted grating lobes and maximize constructive interference.
- **Focal Length Adjustment:** The air gap (height) between the primary radiating patch and the SRR array was fine-tuned to ensure the system operated at the optimal electromagnetic coupling phase. The height of the superstrate from the patch in 2X2 configuration was 11mm whereas in the 4X4 configuration it was 40mm.
- **Structural Consistency:** The fundamental unit cell geometry, including the 2 mm SRR width and the 1.6 mm FR-4 thickness, was maintained to ensure a consistent comparison with the midsem baseline.

RESULTS AND ANALYSIS

2x2 SRR UNIT CELL ARRAY

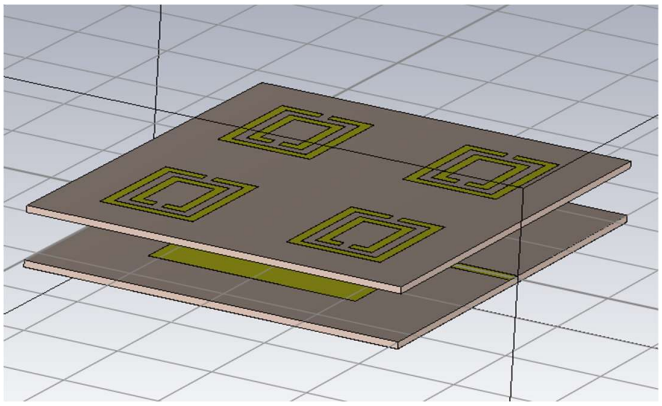


Fig.7.1: Design of 2X2 Metasurface Antenna

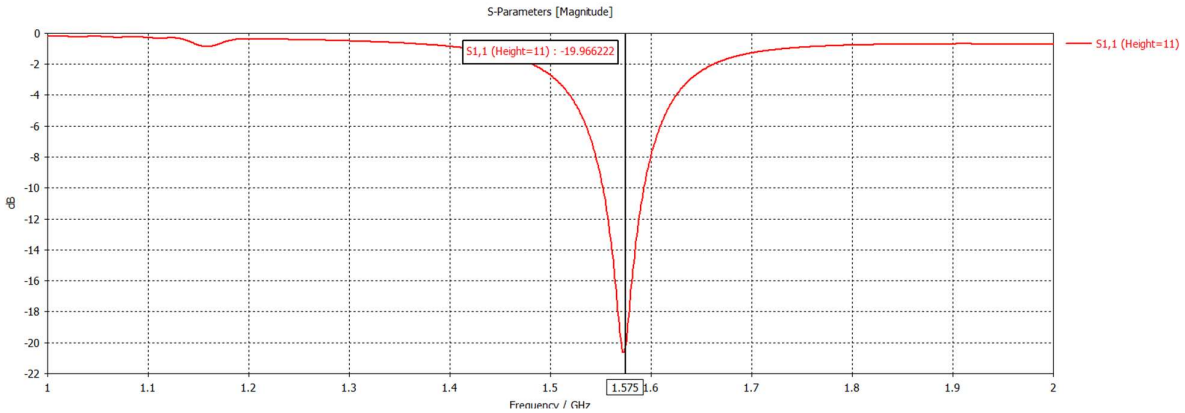


Fig.7.2: S_{11} Parameter

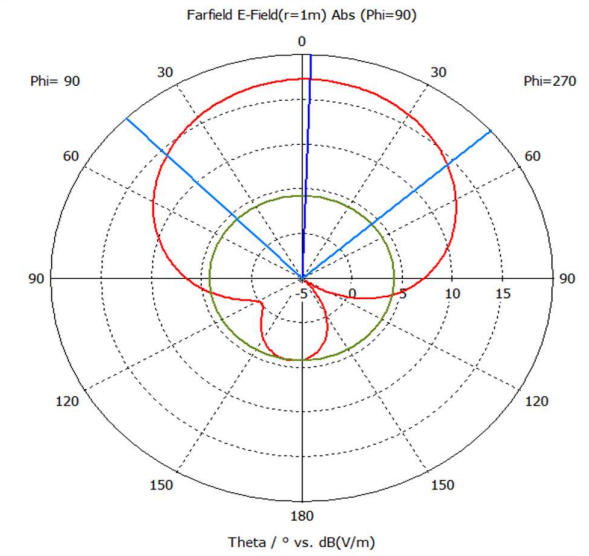


Fig.7.3: E-Field

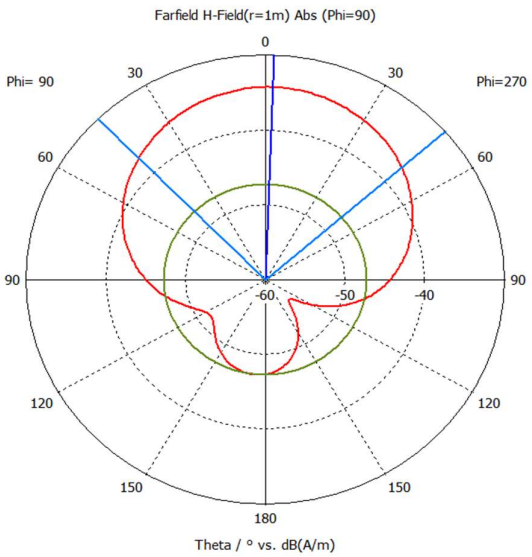


Fig.7.4: H-Field

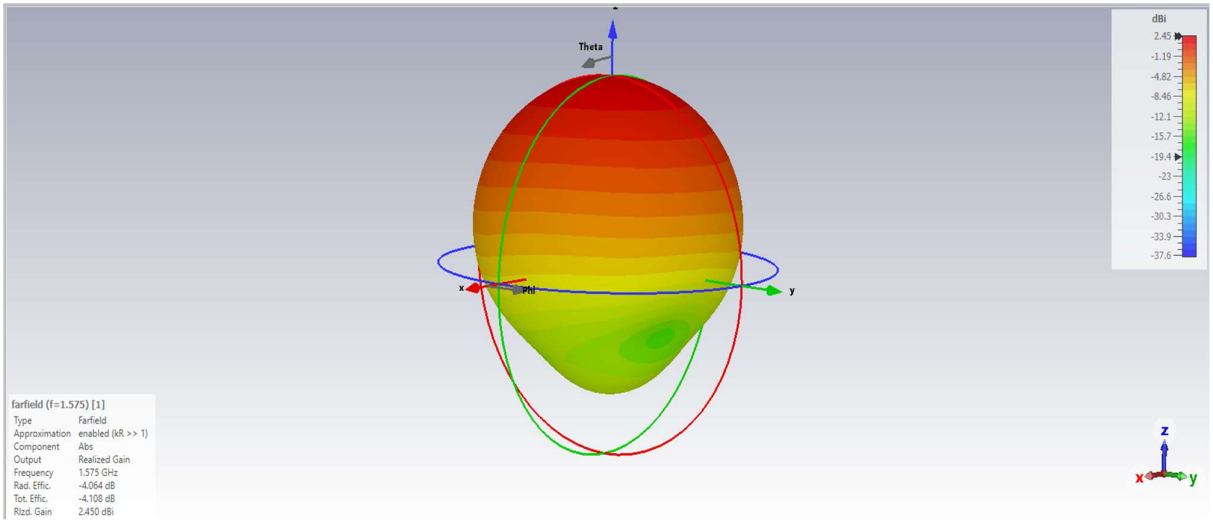


Fig.7.5: Realized Gain of the Antenna (2.45dB)

4x4 SRR UNIT CELL ARRAY

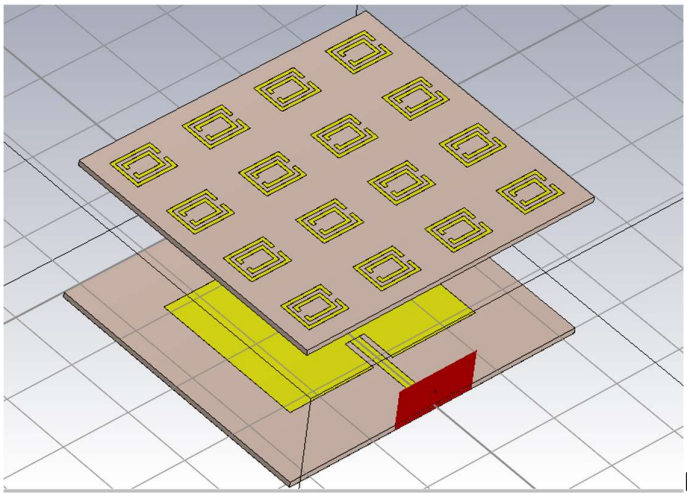


Fig.7.6: Design of 4X4 Metasurface Antenna

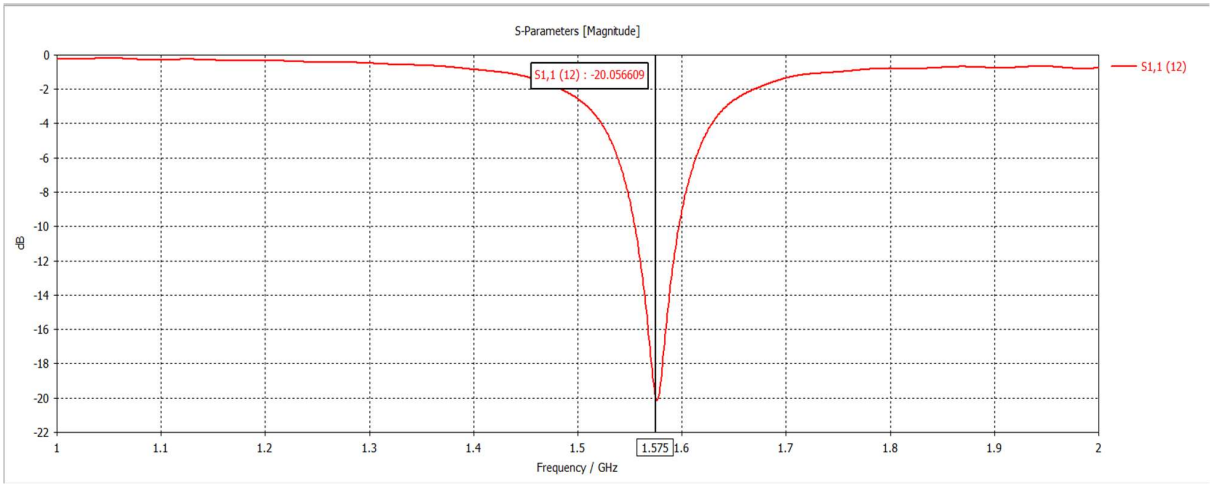


Fig.7.7: S_{11} Parameter

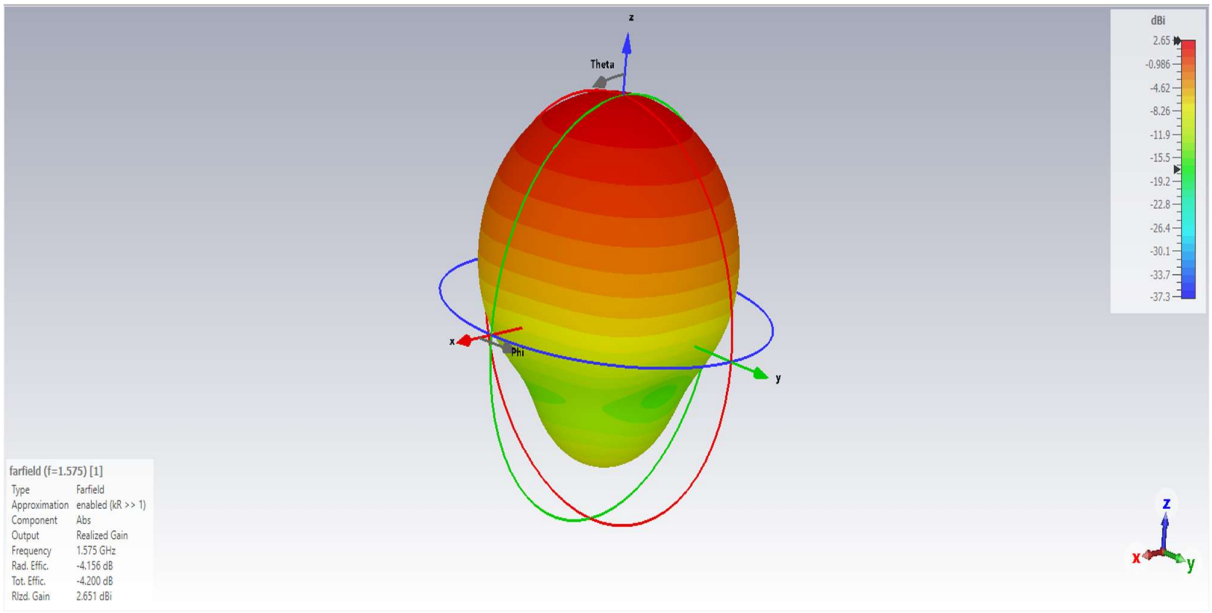


Fig.7.8: Realized Gain of the Antenna (2.65dB)

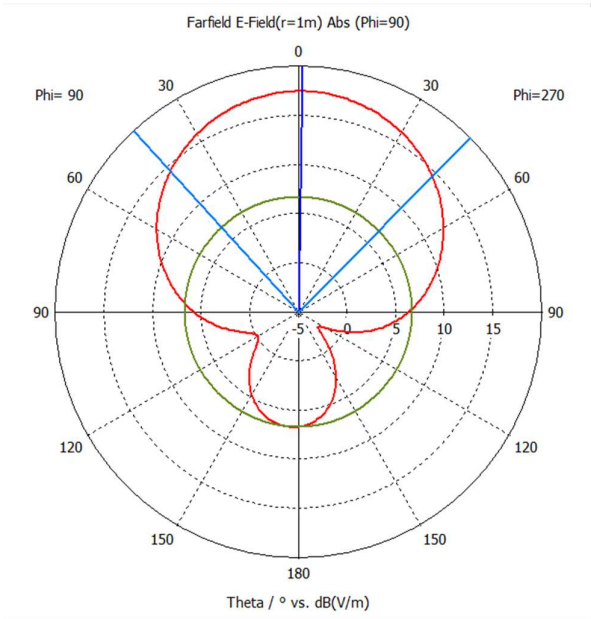


Fig.7.9: E-Field

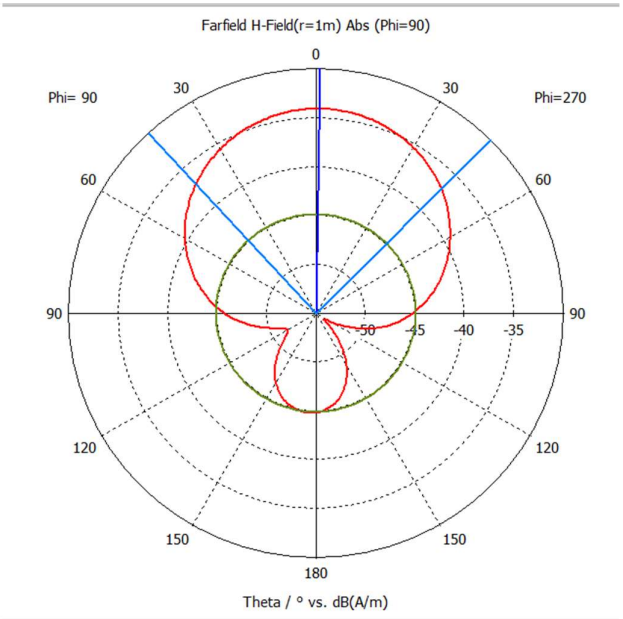


Fig.7.10: H-Field

Comparision of Results:

Design Phase	Configuration	Peak Gain (dB)	Gain Improvement	S11 (at 1.575 GHz)
Baseline	**No Superstrate**	1.93 dB	-	-22.41 dB
Midsem	**Single SRR**	2.26 dB	+0.33 dB	-21.80 dB
Post-Midsem	**2x2 SRR Array**	2.45 dB	+0.52 dB	-19.96 dB
Post-Midsem	**4x4 SRR Array**	2.65 dB	+0.72 dB	-20.05 dB

Electromagnetic Mechanisms of the 4x4 Array

The final 4x4 array implementation addressed the limitations of compact antennas through several specific mechanisms:

- **Phase Front Transformation:** The superstrate modified the near-field phase distribution, transforming the naturally divergent spherical wave front into a focused, uniform planar wave.
- **Transverse Field Cancellation:** Coupling between the patch and the array induced opposing surface currents within the rings, suppressing transverse electric fields and forcing energy to radiate in the broadside direction.
- **Metamaterial Resonance:** The split rings exhibit localized resonance at 1.575 GHz, slowing the phase velocity of the passing wave to create the "lens" focusing effect.

CONCLUSION

This project successfully demonstrated the design, simulation, and optimization of a high-gain, compact embedded GNSS antenna specifically targeting the L1 frequency band (1.575 GHz) for space-constrained IoT applications. The primary research objective was to overcome the inherent low-gain bottleneck typically associated with miniaturized printed antennas on cost-effective substrates like FR-4. By leveraging metamaterial-inspired performance enhancement technique, specifically the transition from a single parasitic director to a periodic 4×4 metasurface superstrate, the antenna's radiation characteristics were significantly improved without increasing its primary physical footprint.

The iterative design process conducted in CST Studio Suite provided a clear quantitative validation of the gain enhancement strategy. The results of each phase are summarized below:

- Baseline Antenna: Established a fundamental resonance at 1.575 GHz with a return loss (S_{11}) of -22.414 dB and a peak realized gain of 1.93 dB.
- Single SRR Superstrate: Validated the concept of the electromagnetic lens, maintaining excellent impedance matching (S_{11} = -21.8 dB) while elevating the peak gain to 2.26 dB.
- 2x2 SRR Array: Successfully implemented the first stage of periodic scaling, increasing the gain further to 2.45 dB with an S_{11} of -19.96 dB.
- 4x4 SRR Array: Achieved the final optimized peak gain of 2.65 dB, representing a total enhancement of 0.72 dB over the baseline model while maintaining stable performance (S_{11} = -20.05 dB).

The efficacy of the 4×4 metasurface layer is attributed to several critical electromagnetic mechanisms. First, the superstrate functions as a phase-front transformer, converting the broad, divergent spherical wave front typical of compact patches into a more uniform planar wave. Second, the electromagnetic coupling between the primary patch and the array induces opposing surface currents that suppress transverse electric fields, thereby focusing the radiated energy in the broadside direction. Finally, the periodic arrangement artificially enlarges the effective physical aperture, capturing and redirecting peripheral near-field energy that would otherwise be lost to surface waves or side lobes.

In conclusion, this research validates that metamaterial-inspired periodic superstrates are a highly viable and efficient strategy for developing high-gain, compact antennas. By engineering the near-field phase distribution, this design achieves superior directivity and signal fidelity, providing a robust solution for the next generation of embedded GNSS receivers in miniaturized electronic devices.

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